

IEEE 802.11ac: Next-generation Standard for Wireless Fidelity

IEEE 802.11ac, the new WLAN standard, promises lower latency over the network as well as lower power consumption. Though it is yet to be ratified by the Wi-Fi Alliance and IEEE, we have already begun to see products based on this standard and also design and test solutions available



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802.11ac, the next step for Wi-Fi 802.11, will breach the barrier of gigabit capacity. Couple this with low latency and an improved power profile and you get the next must-have feature for your gadget.

The IEEE 802.11ac standard has been developed to provide high-throughput wireless local-area network (WLAN). It promises gigabit wireless speeds to consumers.

The Initial Technical Specification Draft 0.1 was confirmed by the IEEE 802.11 Task Group ac (TGac) last year while Wi-Fi Alliance ratification is expected by the end of this year. Though the 802.11ac standard is in draft form

and yet to be ratified by the Wi-Fi Alliance and IEEE, we have already begun to see products available in the market.

What is 802.11?

The IEEE 802.11 is a set of standards used for setting up WLAN between devices. The standards use 2.4, 3.6 and 5GHz frequency bands for data transmission. The most popular protocols are 802.11b and 802.11g, both of which use the 2.4GHz band. These standards use spread-spectrum and frequency-division multiplexing to control interference from microwave ovens and cordless phones.

802.11ac characteristics

WLAN 802.11ac uses a variety of new methods to achieve the tremendous in-

crease in performance to theoretically hit the gigabit capacity and provide high throughput, such as:

1. 6GHz band
2. High-density modulation up to 256 QAM
3. Wider bandwidths through two 80MHz channels or one 160MHz channel
4. Up to eight multiple-input multiple-output (MIMO) spatial streams
5. Multi-user MIMO (MU-MIMO)

The lower power consumption of 802.11ac will be a boon for the multitude of mobile devices that are about to flood the market.

The advanced methods utilised in WLAN 802.11ac present new challenges to design engineers working with this standard. Next, we discuss these challenges and solutions available that help in designing new products based on this standard.

256 QAM and higher frequency band

Quadrature amplitude modulation (QAM) scheme is used to convey a message signal inside two carrier signals so that they can be physically transmitted with better throughput. The two carriers are out of phase with each other by 90°, which makes them quadrature signals. 256 QAM is used because a higher-order QAM makes it possible to transmit more bits per symbol.

The higher-order 256 QAM requires lower error-vector magnitude (EVM) or constellation error in both the transmitter and the receiver as the constellation points are closer together. EVM problems may be caused by car-